

CHEMISTRY | UNIT 02 | LESSON 03

Isotopes, Atomic Mass, and Atomic Identity

Checkpoint Submission

Student name:		Date:	
MLA standard:	MLA.CHEM.ATO.02	Submission:	Checkpoint Submission

Directions

Complete independently. Show every calculation and upload this PDF to U2 L3 Checkpoint Submission.

Part A - Isotope notation and identity

Species	Atomic #	Mass #	p+	n0	e-
chlorine-35	17	35			17
chlorine-37	17	37			17
magnesium-24 ion (+2)	12	24			

Explain why chlorine-35 and chlorine-37 are the same element but different isotopes.

Part B - Weighted average atomic mass

A sample contains 75% isotope X-10 and 25% isotope X-11. Calculate the average atomic mass. Show fraction x mass for each isotope, add the contributions, and include the unit amu.

Isotope	Percent	Decimal fraction	Mass	Contribution
X-10	75%		10 amu	
X-11	25%		11 amu	
Total				

Explain why the result is closer to 10 amu than 11 amu.
